

FINAL EXPLANATION: CHEMISTRY GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: CHEMISTRY GRADE 10 – SUB-STRAND 1.4 CHEMICAL BONDING: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the unit. Think back to the Carbon Paradox from Lesson 1: a diamond chip from a Nairobi jeweller's cutting tool and a pencil graphite stick, both labelled '100% Carbon'. Graphite smeared a grey mark on the ceramic tile and lit the bulb, while diamond scratched the tile and did not conduct. Answer each section in the blank space under the prompt. Write in full sentences, use words such as valence electrons, covalent bond, delocalised electron, Van der Waals forces, ionic, metallic, lattice and structure, and support every claim with evidence from the lessons (use Claim, Evidence, Reasoning where you can). You may draw labelled diagrams. Where a section asks you to use the table, read it carefully first.

Section 1: Valence Electrons and Bond Type

Carbon has 6 electrons in total, but only some of them take part in bonding.

(a) Which electrons take part in bonding, and how many does carbon have? How would you show them in a Lewis dot diagram?

(b) Explain how the octet rule helps you predict the number of bonds carbon forms.

(c) Atoms can gain, lose or share electrons. Explain how valence electrons decide whether a bond is ionic, covalent or metallic, using NaCl and a covalent molecule such as CH₄ as examples.

(a) Valence electrons are the electrons in the outermost shell. Carbon has 4 valence electrons, drawn as 4 unpaired dots around the symbol C. (b) Atoms are most stable with a full outer shell (an octet, or a duplet for Period 1). Carbon has 4 valence electrons, so it needs 4 more to complete its octet, and it does this by forming 4 bonds. (c) In NaCl, sodium (a metal) transfers its 1 valence electron to chlorine (a non-metal). Both then have complete outer shells. This gives Na⁺ and Cl⁻ ions held together by strong electrostatic attraction, which is ionic bonding. In CH₄, carbon and hydrogen atoms share pairs of electrons, so each carbon shares 4 pairs with 4 hydrogens and all atoms reach a noble gas arrangement. This is covalent bonding. In metals, the valence electrons become delocalised and move freely around positive ions, which is metallic bonding. So the number of valence electrons and whether atoms transfer, share or delocalise them decides the bond type.

Section 2: Explaining the Carbon Paradox

Diamond and graphite are both made only of carbon atoms.

(a) In diamond, each carbon atom forms 4 covalent bonds to 4 other carbon atoms in a 3-D tetrahedral network (bond angle 109.5°). In graphite, each carbon atom forms only 3 covalent bonds, in flat hexagonal layers (bond angle 120°). (b) In diamond, all

(a) Describe how carbon atoms are bonded in diamond and how they are bonded in graphite. Include the number of bonds per carbon atom and the shape of the arrangement. (b) Use your answer to (a) to explain why graphite conducts electricity but diamond does not. (c) Explain why diamond scratched the ceramic tile while graphite smeared on it. Your answer must mention the forces inside the structure.	4 valence electrons of every carbon are used in bonds, so there are no free electrons to carry charge and diamond is an insulator. In graphite, the fourth electron of each carbon is not used in a bond and is delocalised, free to move along the layer, so it carries charge and the bulb lit. (c) In diamond, strong covalent bonds run in all three dimensions with no weak links, so it is rigid and very hard and can scratch the tile. In graphite, the strong covalent bonds are only within each layer; the layers are held to each other by weak Van der Waals forces, so the layers slide easily over one another and leave a grey mark on the tile, which is why graphite is soft and can write on paper.
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Section 3: Intermolecular Forces and Metallic Bonding

(a) Water boils at 100 °C while H₂S boils at -60 °C, even though H₂S is a larger molecule. What does this show about the forces between water molecules, and what is the cause? (b) Iodine changes directly from solid to purple vapour when warmed gently. Name the forces holding iodine molecules together and explain why a non-polar substance still has these forces. (c) Describe the bonding in aluminium foil and use it to explain why aluminium conducts electricity and can be bent without breaking.	(a) Water has much stronger attractions between its molecules than H₂S. The cause is hydrogen bonding: the δ^+ hydrogen of one water molecule is attracted to a lone pair on the δ^- oxygen of a neighbouring molecule. Hydrogen bonding needs H bonded to N, O or F, so more energy is needed to separate water molecules and its boiling point is high. (b) Iodine molecules are held by Van der Waals forces, specifically London dispersion forces. Even a non-polar molecule has temporary dipoles as its electron cloud moves, and in large iodine molecules with big, polarisable electron clouds these attractions are fairly strong yet weak enough that iodine changes easily to vapour. (c) Aluminium has a regular lattice of positive ions surrounded by a sea of delocalised electrons that belong to no single ion; the electrostatic attraction between them is metallic bonding. The free electrons move and carry charge, so aluminium conducts. The bonding is non-directional, so layers of ions can slide over each other without breaking bonds, so the metal is malleable and ductile.
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Section 4: Bond Type, Structure and Properties

Use the table of results from your investigation to answer the questions below.	(a) In solid NaCl the Na⁺ and Cl⁻ ions are locked in fixed positions in the lattice and cannot move. When dissolved, the ions are separated and free to move, so they carry charge. (b) Sugar is a simple molecular substance made of neutral molecules. It dissolves but forms no ions and has no free electrons, so there are no charged particles to carry current. (c) In SiO₂ all the valence electrons are used in covalent bonds in a continuous network, so there are no free particles. In graphite each carbon forms only 3
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Su bst an ce	Str uct ure typ e	Dis sol ve s in wa ter	Co nd uct s as sol id	Co nd uct s wh en dis sol ve d
Na Cl	Gia nt ioni c	Ye s	No	Ye s
Su gar	Si mp le mo lec ula r	Ye s	No	No
Si O ₂ (sa nd)	Gia nt cov ale nt	No	No	Not ap plic abl e
Alu mi niu m	Gia nt me talli c	No	Ye s	Not ap plic abl e
Gr ap hit e	Gia nt cov ale nt	No	Ye s	Not ap plic abl e

(a) Explain why NaCl conducts only when dissolved, but not as

bonds, leaving one delocalised electron per atom that can move through the layer, so graphite conducts like a metal even though it is a covalent structure. (d) Simple molecular substances, such as sugar, iodine and water, have the lowest melting points because only weak intermolecular forces (Van der Waals forces or hydrogen bonds) need to be overcome and not the strong bonds throughout a giant structure.

a solid. (b) Why does sugar not conduct, even when dissolved? (c) Graphite and SiO₂ are both giant covalent, yet only graphite conducts. Explain this. (d) Which type of structure would you expect to have the lowest melting points, and why?	
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Section 5: From Structure to Use	
Answer the driving question: Why do diamond and graphite, both made only of carbon atoms, have completely different properties, and how does the way atoms bond together determine what a substance looks like, how it behaves, and what we can use it for? (a) Write a Claim, Evidence and Reasoning paragraph that answers the driving question using diamond and graphite. (b) Give one use of diamond and one use of graphite in Kenya and link each use to its structure. (c) A student suggests that a copper wire or an NaCl solution could replace graphite in a dry-cell battery. Choose one and evaluate this idea using bonding ideas.	(a) Claim: Diamond and graphite differ because their carbon atoms are bonded and arranged differently, not because they contain different atoms. Evidence: Diamond scratched the tile and did not conduct; graphite smeared on the tile and lit the bulb. Reasoning: In diamond each carbon forms 4 covalent bonds in a rigid 3-D tetrahedral network with no free electrons, so it is very hard and an insulator. In graphite each carbon forms 3 covalent bonds in hexagonal layers, one electron per atom is delocalised so it conducts, and the layers are held only by weak Van der Waals forces so they slide and graphite is soft. So the bond type determines the structure, the structure determines the properties, and the properties determine the uses. (b) Diamond is used in jewellers' cutting tools in Nairobi because its network of strong covalent bonds in all directions makes it extremely hard. Graphite is used in pencils because its layers slide and leave marks, and in dry-cell batteries because its delocalised electrons conduct electricity. (c) A copper wire could conduct because its sea of delocalised electrons carries charge, so copper would work as a conductor, although a metal is not the same as graphite as an electrode material. An NaCl solution conducts through mobile ions, but a liquid is not suitable as a solid electrode. Both are conductors, but the structure of graphite, a solid that conducts through delocalised electrons, is why it is chosen for the battery.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)

Use of valence electrons and bond types	Accurately explains how valence electrons and the octet rule lead to ionic, covalent and metallic bonding, with correct examples such as NaCl and CH ₄ .	Identifies valence electrons and the three bond types with mostly correct examples but limited explanation of why each forms.	Confuses bond types or valence electrons, or gives examples without explanation.
Explanation of diamond and graphite	Clearly links number of bonds, geometry, delocalised electrons and Van der Waals forces to conductivity and hardness of both substances.	Explains most differences between diamond and graphite but misses one link, such as the weak forces between layers.	States that the properties differ without linking them to the bonding or structure.
Use of evidence and data	Uses the table and lesson observations accurately to support every claim, including the graphite anomaly.	Uses some evidence correctly but some claims are unsupported.	Gives little or incorrect evidence.
Structure-property-use reasoning	Connects bond type to structure, properties and a real Kenyan use, and applies the ideas to a new situation.	Connects structure to properties and uses, with limited application to a new situation.	Lists uses with no reasoning from structure.
Scientific language and communication	Uses correct terms precisely and organises answers in clear Claim, Evidence, Reasoning form.	Uses mostly correct terms with generally clear organisation.	Uses few scientific terms or the answers are unclear.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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